TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	89.0 ± 1.26
Correlation Stability	89.4 ± 1.02
Distribution Consistency	91.7 ± 1.42
Entropy Preservation	67.6 ± 1.02
Mean Z Score Drift	0.1 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	91.2 ± 1.60
Feature Accuracy	90.4 ± 1.36
Novelty Score	0.9 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: chronic_kidney_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	400
Columns	25
Numeric features	25
Categorical features	0
Missing cells (%)	10.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	51.483	51.991	+0.507	1.0%	OK
bp	76.469	76.469	-0.000	0.0%	OK
sg	1.017	1.017	-0.001	0.1%	OK
al	1.017	0.938	-0.079	7.8%	OK
su	0.450	0.406	-0.044	9.8%	OK
rbc	0.190	0.119	-0.071	37.3%	Drift
pc	0.227	0.169	-0.058	25.6%	Drift
pcc	0.106	0.084	-0.022	20.4%	Warn
ba	0.056	0.031	-0.024	43.7%	Drift
bgr	148.037	143.481	-4.555	3.1%	OK
bu	57.426	58.210	+0.784	1.4%	OK
sc	3.072	3.881	+0.808	26.3%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	89.0 ± 1.26
Correlation Stability (*2)	89.4 ± 1.02
Distribution Consistency (*3)	91.7 ± 1.42
Entropy Preservation (*4)	67.6 ± 1.02
Mean Z Score Drift (*5)	0.1 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	91.2 ± 1.60
Feature Accuracy (*10)	90.4 ± 1.36
Novelty Score (*11)	0.9 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6024999999999999	0.6024999999999999	0.6024999999999999	0	
F1	0.7507214882964476	0.7507214882964476	0.7507214882964476	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9825000000000002	0.9625	0.99375	0.011250000000000015	
F1	0.9848619694663338	0.9684379862730303	0.9948415615895614	0.009979592123227665	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.99875	0.9762500000000001	0.9962500000000001	-0.002499999999999991	
F1	0.9989247311827956	0.979655503781941	0.9968411717687967	-0.002083559413998914 6	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6125	0.6075	0.6475000000000001	0.03500000000000002	
F1	0.5885198185573248	0.6914031515129024	0.6646735862850531	0.0761537677277282	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7225	0.7425	0.6412500000000001	-0.08125	
F1	0.7118458393632935	0.817953764187169	0.7052282549111674	-0.006617584452126624	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.95	0.9037499999999999	0.9337500000000001	-0.016249999999999987	
F1	0.9513543755805717	0.9084576617907569	0.9371357758524615	-0.014218599728110116	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9862500000000001	0.9775	0.9949999999999999	0.008750000000000013	
F1	0.9880875040761434	0.9818963536564975	0.9957205669259999	0.007633062849856409	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9525	0.9199999999999999	0.9362499999999999	-0.016249999999999987	*
F1	0.9587663012688215	0.9284147254299178	0.943417389324139	-0.015348911944682598	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9100000000000001	0.9012500000000001	0.90625	-0.003749999999999977	
F1	0.92384446063128	0.917888015261118	0.9210331753941201	-0.002811285237159943	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.97875	0.98375	0.9912500000000002	0.012499999999999989	
F1	0.9805471229310466	0.9864319613605957	0.9925785154891436	0.012031392558097026	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	48.0	80.0	1.02	1.0	0.0	nan	0.0	0.0	0.0	121.0
2	7.0	50.0	1.02	4.0	0.0	nan	0.0	0.0	0.0	nan
3	62.0	80.0	1.01	2.0	3.0	0.0	0.0	0.0	0.0	423.0
4	48.0	70.0	1.005	4.0	0.0	0.0	1.0	1.0	0.0	117.0
5	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
6	60.0	90.0	1.015	3.0	0.0	nan	nan	0.0	0.0	74.0
7	68.0	70.0	1.01	0.0	0.0	nan	0.0	0.0	0.0	100.0
8	24.0	nan	1.015	2.0	4.0	0.0	1.0	0.0	0.0	410.0
9	52.0	100.0	1.015	3.0	0.0	0.0	1.0	1.0	0.0	138.0
10	53.0	90.0	1.02	2.0	0.0	1.0	1.0	1.0	0.0	70.0
11	50.0	60.0	1.01	2.0	4.0	nan	1.0	1.0	0.0	490.0
12	63.0	70.0	1.01	3.0	0.0	1.0	1.0	1.0	0.0	380.0
13	68.0	70.0	1.015	3.0	1.0	nan	0.0	1.0	0.0	208.0
14	68.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	98.0
15	68.0	80.0	1.01	3.0	2.0	0.0	1.0	1.0	1.0	157.0
16	40.0	80.0	1.015	3.0	0.0	nan	0.0	0.0	0.0	76.0
17	47.0	70.0	1.015	2.0	0.0	nan	0.0	0.0	0.0	99.0
18	47.0	80.0	nan	nan	nan	nan	nan	0.0	0.0	114.0
19	60.0	100.0	1.025	0.0	3.0	nan	0.0	0.0	0.0	263.0
20	62.0	60.0	1.015	1.0	0.0	nan	1.0	1.0	0.0	100.0

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	58.0	90.0	1.01236977136112 64	3.0	0.0	0.0	1.0	1.0	0.0	311.0
2	23.0	70.0	1.01739305646048 65	4.0	0.0	0.0	0.0	0.0	0.0	150.0
3	42.0	60.0	1.02033961428687 24	0.0	0.0	0.0	0.0	0.0	0.0	92.0
4	43.0	70.0	1.01200834298945 9	0.0	0.0	0.0	0.0	0.0	0.0	111.0
5	27.0	70.0	1.02104926136877 03	0.0	0.0	0.0	0.0	0.0	0.0	145.0
6	61.0	70.0	1.02576727736979 88	0.0	0.0	0.0	0.0	0.0	0.0	100.0
7	35.0	80.0	1.005	3.0	0.0	1.0	0.0	0.0	0.0	nan
8	30.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	96.0
9	52.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	99.0
10	65.0	80.0	1.015	0.0	0.0	nan	0.0	0.0	0.0	115.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	60.0	60.0	1.01366338914578 96	3.0	1.0	0.0	1.0	1.0	0.0	284.0
2	24.0	70.0	1.01461652531317 85	0.0	0.0	0.0	0.0	0.0	0.0	345.0
3	54.0	110.0	1.00988547612603	0.0	0.0	1.0	0.0	1.0	1.0	122.0
4	35.0	80.0	1.01719979327700 72	1.0	0.0	0.0	0.0	0.0	0.0	328.0
5	73.0	100.0	1.01620033800409 25	2.0	0.0	0.0	0.0	1.0	0.0	142.0
6	60.0	90.0	1.01070403577618 12	4.0	1.0	1.0	1.0	0.0	0.0	302.0
7	64.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	113.0
8	42.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	98.0
9	12.0	60.0	1.015	3.0	0.0	1.0	1.0	1.0	0.0	nan
10	23.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	111.0

TabuLLM: AI Benchmark Report

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	69.0	70.0	1.01054812607053 47	2.0	0.0	0.0	1.0	0.0	1.0	119.0
2	53.0	110.0	1.01037106254685 16	1.0	0.0	0.0	0.0	0.0	0.0	42.0
3	58.0	110.0	1.01982878904385 9	4.0	4.0	1.0	1.0	1.0	0.0	180.0
4	55.0	80.0	1.01989386664942 8	0.0	0.0	0.0	0.0	0.0	0.0	251.0
5	45.0	70.0	1.01956359419919 74	0.0	0.0	0.0	0.0	0.0	0.0	50.0
6	56.0	80.0	1.02284408741087 97	0.0	0.0	0.0	0.0	0.0	0.0	86.0
7	34.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	121.0
8	33.0	60.0	nan	nan	nan	0.0	0.0	0.0	0.0	130.0
9	60.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	124.0
10	57.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	132.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	47.0	70.0	1.02041176573847 68	0.0	0.0	0.0	0.0	0.0	0.0	137.0
2	58.0	80.0	1.02218258851399 5	0.0	0.0	0.0	0.0	0.0	0.0	90.0
3	43.0	80.0	1.02104121494583 34	1.0	0.0	0.0	0.0	0.0	0.0	140.0
4	37.0	80.0	1.02104759990703 8	0.0	0.0	0.0	0.0	0.0	0.0	115.0
5	39.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	115.0
6	78.0	80.0	1.01795556054950 63	2.0	0.0	0.0	0.0	0.0	0.0	114.0
7	44.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0
8	55.0	90.0	1.01	2.0	1.0	1.0	1.0	0.0	0.0	273.0
9	45.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	360.0
10	72.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	308.0

TabuLLM: AI Benchmark Report

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	21.0	80.0	1.0153070938297097	4.0	0.0	0.0	0.0	0.0	0.0	114.0
2	86.0	60.0	1.0174676942231056	0.0	0.0	0.0	0.0	0.0	0.0	180.0
3	62.0	70.0	1.0096200111122422	0.0	0.0	0.0	0.0	1.0	0.0	147.0
4	30.0	90.0	1.0213394240249438	0.0	0.0	0.0	0.0	0.0	0.0	112.0
5	33.0	60.0	1.0208528680201754	0.0	0.0	0.0	0.0	0.0	0.0	80.0
6	39.0	80.0	1.0270342655404303	0.0	0.0	0.0	0.0	0.0	0.0	114.0
7	41.0	80.0	1.015	1.0	4.0	1.0	0.0	0.0	0.0	210.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	87.0
9	26.0	70.0	1.015	0.0	4.0	nan	0.0	0.0	0.0	250.0
10	50.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	137.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	79.0	100.0	1.0172989631542542	1.0	0.0	1.0	0.0	0.0	0.0	163.0
2	20.0	60.0	1.0132523627494041	0.0	0.0	0.0	0.0	0.0	0.0	94.0
3	41.0	70.0	1.0203888641412568	0.0	0.0	0.0	0.0	0.0	0.0	290.0
4	67.0	100.0	1.0184927105803685	3.0	4.0	0.0	0.0	0.0	0.0	269.0
5	43.0	70.0	1.0176548889108175	2.0	0.0	0.0	0.0	0.0	0.0	137.0
6	55.0	80.0	1.0182487449434949	0.0	0.0	0.0	0.0	0.0	0.0	86.0
7	8.0	60.0	1.025	3.0	0.0	0.0	0.0	0.0	0.0	78.0
8	60.0	70.0	1.01	1.0	0.0	nan	0.0	0.0	0.0	109.0
9	nan	70.0	1.01	3.0	0.0	0.0	0.0	0.0	0.0	110.0
10	62.0	90.0	1.02	2.0	1.0	nan	0.0	0.0	0.0	169.0

TabuLLM: AI Benchmark Report

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	72.0	80.0	1.013198594956849	2.0	2.0	0.0	0.0	0.0	0.0	79.0
2	54.0	70.0	1.022910278185663 4	0.0	0.0	0.0	0.0	0.0	0.0	106.0
3	34.0	80.0	1.017476873047030 3	0.0	0.0	0.0	0.0	0.0	0.0	111.0
4	43.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	126.0
5	59.0	90.0	1.021007520109347 4	2.0	0.0	0.0	0.0	0.0	0.0	197.0
6	62.0	80.0	1.015331931563979 5	4.0	1.0	0.0	1.0	0.0	0.0	194.0
7	nan	70.0	1.02	0.0	0.0	nan	nan	0.0	0.0	219.0
8	29.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	127.0
9	50.0	90.0	1.015	1.0	0.0	1.0	1.0	0.0	0.0	nan
10	65.0	70.0	1.02	1.0	0.0	1.0	1.0	0.0	0.0	139.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	61.0	90.0	1.013351953484012 7	2.0	3.0	0.0	1.0	0.0	0.0	347.0
2	40.0	80.0	1.017896492216833 2	0.0	0.0	0.0	0.0	0.0	0.0	95.0
3	66.0	90.0	1.019750577849256 8	0.0	3.0	1.0	1.0	0.0	0.0	256.0
4	61.0	70.0	1.014291391908522 1	3.0	1.0	0.0	1.0	1.0	0.0	237.0
5	46.0	80.0	1.022694566491624 7	0.0	0.0	0.0	0.0	0.0	0.0	101.0
6	26.0	70.0	1.011938071721156 8	0.0	0.0	0.0	0.0	0.0	0.0	106.0
7	53.0	100.0	1.01	1.0	3.0	1.0	0.0	0.0	0.0	213.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	nan
9	45.0	80.0	1.02	3.0	0.0	0.0	1.0	0.0	0.0	425.0
10	33.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	89.0

TabuLLM: AI Benchmark Report

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	22.0	70.0	1.016052727118967	3.0	2.0	1.0	0.0	0.0	0.0	412.0
2	39.0	100.0	1.0022574660702075	4.0	2.0	1.0	1.0	0.0	0.0	217.0
3	42.0	80.0	1.017767120264699	0.0	0.0	0.0	1.0	1.0	0.0	91.0
4	58.0	100.0	1.020394991877763	3.0	0.0	0.0	0.0	0.0	0.0	59.0
5	71.0	90.0	1.014650389703987	4.0	0.0	0.0	1.0	0.0	0.0	256.0
6	57.0	70.0	1.015149636269486	2.0	2.0	0.0	0.0	0.0	0.0	204.0
7	46.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	102.0
8	22.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	97.0
9	65.0	80.0	1.01	3.0	3.0	nan	0.0	0.0	0.0	294.0
10	23.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	45.0	100.0	1.011035380286482	4.0	2.0	0.0	0.0	0.0	0.0	138.0
2	21.0	80.0	1.013906132928782	1.0	0.0	0.0	0.0	0.0	0.0	188.0
3	51.0	70.0	1.017757475144913	0.0	0.0	0.0	1.0	0.0	0.0	105.0
4	62.0	80.0	1.027956912937534	0.0	0.0	0.0	0.0	0.0	0.0	99.0
5	66.0	70.0	1.022563375813854	4.0	2.0	0.0	1.0	0.0	0.0	206.0
6	39.0	80.0	1.014421215681293	1.0	0.0	0.0	0.0	0.0	0.0	114.0
7	11.0	80.0	1.01	3.0	0.0	nan	0.0	0.0	0.0	nan
8	61.0	90.0	1.015	0.0	2.0	nan	0.0	0.0	0.0	nan
9	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
10	70.0	60.0	nan	nan	nan	nan	nan	0.0	0.0	120.0